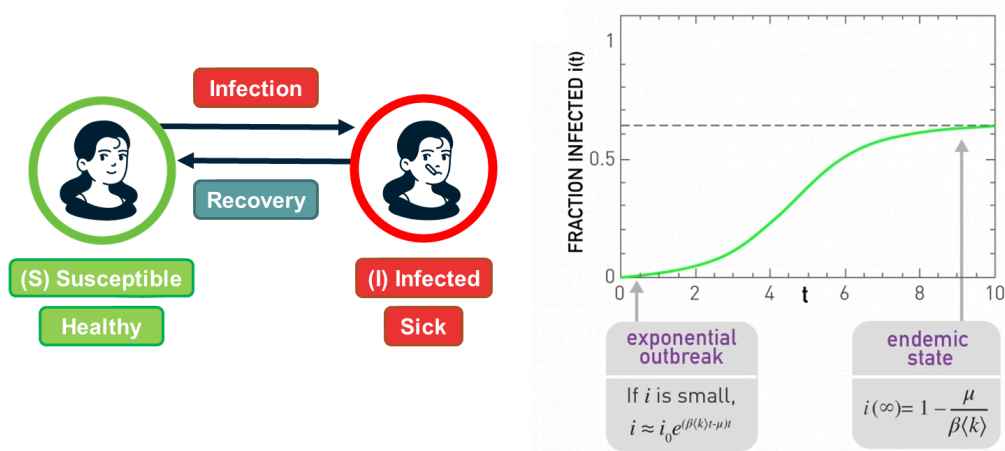


L9: SIS Model



(<https://gatech.instructure.com/courses/552990/files/73570101/download?wrap=1>)

Figure 10.5 from Network Science by Albert-László Barabási

The SI model is unrealistic because it assumes that an infected individual stays infected. In practice, thanks to our immune system, we can recover from most infections after some time period. In the SIS model, we extend the SI model with an additional transition, from the I back to the S state to capture this recovery process.

The recovery of an infected individual is also a probabilistic process. As we did with the infection process, let us define as μdt the probability that an infected individual recovers during an infinitesimal time period dt . If the density of infected individuals is $i(t)$, then the transition rate from the I state to the S state is $\mu i(t)$.

So, the differential equation that describes the SIS model is similar with the SI equation – but with a negative term that decreases the density of infected individuals as follows:

$$\frac{di(t)}{dt} = \bar{k}\beta i(t)(1 - i(t)) - \mu i(t)$$

The initial condition is, again, $i(0) = i_0$.

As in the case of the SI model, this differential equation can be solved despite the quadratic term: $i(t) = (1 - \frac{\mu}{\bar{k}\beta}) \frac{c e^{(\bar{k}\beta - \mu)t}}{1 + c e^{(\bar{k}\beta - \mu)t}}$ (1)

where c is a constant that depends on the initial condition as follows:

$$c = \frac{i_0}{(1 - i_0) - \frac{\mu}{\bar{k}\beta}}$$

Note that if we set $\mu = 0$, we get the same solution we had previously derived for the SI model.

The SIS model can lead to two very different outcomes, depending on the magnitude of the recovery rate μ relative to the cumulative infection rate $\bar{k}\beta$:

If $\bar{k}\beta < \mu$, then the exponent in the previous solution is negative and the density of infected individuals drops exponentially fast from i_0 to zero. In other words, the original infection does not cause an outbreak. This happens when the recovery of the original infected individual takes place faster than the infection of his/her susceptible neighbors.

In the opposite case, when $\bar{k}\beta > \mu$, we have an exponential outbreak for small values of t (when the density of infected individuals is quite smaller than 1). In that regime, we can approximate the solution of the SIS model with the following equation: $i(t) \approx i_0 e^{(\bar{k}\beta - \mu)t}$

The time constant for the SIS model, during that exponential outbreak, is $\frac{1}{\bar{k}\beta - \mu}$

As time increases, when $\bar{k}\beta > \mu$, we get that fraction of infected individuals tends to $1 - \frac{\mu}{\bar{k}\beta}$. In other words, we get a persistent epidemic in which even though individuals keep moving between the S and I states, the percentage of the population that remains sick is practically constant. This is referred to as the “endemic state”.

The ratio $\frac{\bar{k}\beta}{\mu}$ is critical for the SIS model: if it is larger than 1, the SIS model predicts that even a small outbreak will lead to an endemic state. Otherwise, the outbreak will die out. This is why we define that the *epidemic threshold* of this model is equal to one.

Food For Thought

Derive equation (1) in detail.